

# Family Command Center

Complete Project Documentation

Version 1.0.0 | iOS & Android | June 2026

| Field              | Value                                                                |
|--------------------|----------------------------------------------------------------------|
| Bundle ID          | com.familycommandcenter.app                                          |
| Platforms          | iOS + Android (Expo Bare Workflow, New Architecture enabled)         |
| React Native       | 0.79.6 / React 19 / Expo SDK 53 / TypeScript 5.8                     |
| Backend            | Node.js + Express 4 + Prisma 5 + PostgreSQL                          |
| AI Engine          | Google Vertex AI (Gemini) via @google-cloud/vertexai v1.12           |
| Banking            | Plaid API v42.2                                                      |
| Push Notifications | Firebase Admin SDK v12 (FCM)                                         |
| State Management   | Zustand v5 — 43 stores persisted via react-native-mmkv               |
| Deep Link Scheme   | familycommandcenter://                                               |
| Admin Panel        | Next.js web app (family-command-center-admin)                        |
| Child Safety App   | Separate RN app (family-guardian-child) for Android children devices |

## 1. Project Overview

Family Command Center is a comprehensive family management platform that replaces 10+ single-purpose apps with one role-aware hub covering household operations, finances, parental controls, and AI assistance.

### Member Roles & Access

| Role        | Access Level                                                             |
|-------------|--------------------------------------------------------------------------|
| Parent      | Full access — all tabs, all screens, all management tools                |
| Guardian    | Full access — trusted adult with parent-level permissions                |
| Child       | Limited — tasks, calendar, AI chat, mood, habits, kids mode              |
| Grandparent | Read-heavy — legacy vault, timeline, birthdays, journals, AI safety chat |
| Caregiver   | Operational — tasks, calendar, operations read-only, no finance          |

### Tech Stack Summary

| Layer           | Technology                                                             |
|-----------------|------------------------------------------------------------------------|
| Mobile Frontend | React Native 0.79.6 + Expo SDK 53 + TypeScript + React Navigation v6   |
| State / Storage | Zustand v5 (43 stores) + react-native-mmkv (encrypted MMKV)            |
| Backend API     | Node.js + Express 4 + Prisma ORM 5 + PostgreSQL                        |
| AI / Vision     | Google Vertex AI Gemini (chat, receipt scanning, family insights)      |
| Banking         | Plaid API v42 (transactions, balances, autofill detection)             |
| Push / Commands | Firebase Admin SDK FCM (parental commands, geofence alerts, SOS)       |
| Child Safety    | Separate Android app with VPN-based web filtering (native Java/Kotlin) |

## 2. Navigation Structure

| State           | Navigator                    | Description                                                               |
|-----------------|------------------------------|---------------------------------------------------------------------------|
| Unauthenticated | AuthNavigator                | SignIn, SignUp, ForgotPassword                                            |
| First-time user | OnboardingNavigator          | 8 setup screens — gesture-back disabled, skip buttons on optional screens |
| Active user     | MainNavigator + TabNavigator | 5-tab bottom nav with custom animated tab bar                             |

### Tab Structure

| Tab          | Root Screen               | Stack Depth                                             | Role Restriction         |
|--------------|---------------------------|---------------------------------------------------------|--------------------------|
| Home         | DashboardScreen           | CommandWall, Notifications, Search, WeeklyReport        | All roles                |
| Family       | FamilyProfilesScreen      | 35+ screens incl. Guardian sub-stack (7 screens)        | Role-filtered per screen |
| Finance      | FinanceDashboardScreen    | 14 screens — Budgeting through ReceiptScanner           | Parent / Guardian only   |
| Operations   | OperationsDashboardScreen | 20 screens — Vehicles through CarpoolManager            | Parent / Guardian only   |
| AI Assistant | AIAssistantScreen         | 6 screens — memory, coaching, negotiation, twin, safety | Role-filtered            |

### 3. Complete Screen Inventory (100 Screens)

#### Auth & Onboarding (11)

| Screen               | Description                                                                                   |
|----------------------|-----------------------------------------------------------------------------------------------|
| SignInScreen         | Email/password with inline validation, biometric shortcut for returning users                 |
| SignUpScreen         | 3-step registration: personal info, role, security. DOB auto-mask, strong password (8+ chars) |
| ForgotPasswordScreen | Email reset with 60-second resend countdown                                                   |
| SplashScreen         | Animated logo (800ms on return visits, 3.2s first launch)                                     |
| WelcomeScreen        | Feature showcase with demo mode warning                                                       |
| FamilySetupScreen    | Family name, auto-detected timezone, currency — Step 1 of 6                                   |
| MemberSetupScreen    | Add first member with 2x2 role chip grid — Step 2 of 6                                        |
| HomeSetupScreen      | Home address setup — Step 3, skippable                                                        |
| VehicleSetupScreen   | Register vehicles — Step 4, skippable                                                         |
| FinancialSetupScreen | Initial finance config — Step 5, skippable                                                    |
| GoalsSetupScreen     | Family goals — Step 6, skippable                                                              |

#### Dashboard (5)

| Screen              | Description                                                                   |
|---------------------|-------------------------------------------------------------------------------|
| DashboardScreen     | Family hub: member statuses, tasks due today, finance snapshot, quick actions |
| CommandWallScreen   | Pinned family announcements and shared broadcast messages                     |
| NotificationsScreen | All notifications filtered by active member role                              |
| SearchScreen        | Global search across all family data                                          |
| WeeklyReportScreen  | Weekly summary: tasks, spending, mood trends, achievements                    |

## Family Screens (31)

| Screen                   | Description                                                  |
|--------------------------|--------------------------------------------------------------|
| FamilyProfilesScreen     | Member cards with status, points, level                      |
| ProfileSwitcherScreen    | PIN-protected profile switching                              |
| AddMemberScreen          | Add member with role, avatar, permissions, invite            |
| MemberDetailsScreen      | Full member profile: health, permissions, history            |
| FamilyBoardScreen        | Shared bulletin board — posts, photos, links                 |
| CalendarScreen           | Shared family calendar with color-coded member events        |
| TasksScreen              | Role-filtered task list with priority, due dates, recurrence |
| JoinFamilyScreen         | Join existing family via 6-character invite code             |
| FamilyInviteQRScreen     | Generate QR code for family invites                          |
| JoinRequestsScreen       | Review and approve/deny join requests                        |
| HomeworkTrackerScreen    | Track assignments per child by subject and due date          |
| SchoolCenterScreen       | School schedule, grades, events                              |
| LegacyVaultScreen        | Family heirloom documents and stories                        |
| FamilyTimelineScreen     | Chronological family milestones                              |
| BirthdayTrackerScreen    | Birthday countdown with gift ideas                           |
| GiftPlannerScreen        | Track gifts by occasion through purchase lifecycle           |
| FamilyJournalScreen      | Shared and private journal entries with mood tagging         |
| RelationshipHealthScreen | Relationship check-ins and communication score               |
| MoodTrackerScreen        | Daily mood logging per member with trend charts              |
| HabitsScreen             | Build and track personal habits with streaks                 |
| ConflictResolverScreen   | AI-mediated conflict resolution step-by-step                 |
| ReputationScreen         | Points history, tier badges, reputation leaderboard          |
| FamilyMeetingScreen      | Structured meeting agenda and action items                   |
| FamilyPollsScreen        | Create polls and vote — family democracy                     |
| ChoreRotationScreen      | Auto-rotating chore assignments with completion tracking     |
| AllowanceScreen          | Allowance amounts, payment history, chore linkage            |
| FamilyGoalsScreen        | Shared goals with milestones and progress bars               |
| BucketListScreen         | Family bucket list with priority and completion              |
| ActivitiesTrackerScreen  | Extracurricular activities, schedules, costs                 |
| KidsModeScreen           | Simplified large-icon interface for children                 |
| AchievementsScreen       | Earned badges, streaks, achievement history                  |

## Guardian / Parental Control Screens (7)

| Screen                  | Description                                                          |
|-------------------------|----------------------------------------------------------------------|
| GuardianDashboardScreen | Overview: device status, battery, last location, active mode         |
| ChildDeviceDetailScreen | Device details, command history, live GPS, app usage breakdown       |
| GeofenceScreen          | Create/manage zones (Home, School) with entry/exit alerts            |
| ScreenTimeScreen        | Set daily limits per child, blocked apps, scheduled downtime windows |
| SOSAlertsScreen         | View and resolve SOS alerts with GPS location display                |
| ApprovalRequestsScreen  | Approve/deny child requests (app installs, screen time, purchases)   |
| PairDeviceScreen        | Pair new child device using 6-digit pairing code                     |

## Finance Screens (14)

| Screen                 | Description                                                        |
|------------------------|--------------------------------------------------------------------|
| FinanceDashboardScreen | Net worth, monthly spend vs. income, account balances overview     |
| BudgetingScreen        | Category budgets with Plaid autofill suggestions and progress bars |
| BillsScreen            | Recurring bills with Plaid detection and due date alerts           |
| SubscriptionsScreen    | Active subscriptions detected from transactions                    |
| AssetsScreen           | Asset tracking: home, vehicles, investments, valuables             |
| WealthBuilderScreen    | Long-term wealth planning and net worth trends                     |
| InsuranceManagerScreen | Insurance policies with expiry tracking and auto-detection         |
| DebtPayoffScreen       | Snowball/Avalanche planner with Plaid balance sync                 |
| DebtDetailScreen       | Individual debt: payment history, interest paid, payoff timeline   |
| UtilityTrackerScreen   | Utility bills with month-over-month trend charts                   |
| ConnectBankScreen      | Plaid Link bank connection flow (expo-web-browser hosted)          |
| TransactionsScreen     | Full transaction history with category, search, filters            |
| SpendingInsightsScreen | Category breakdown and 6-month monthly trend analysis              |
| ReceiptScannerScreen   | Camera receipt scanning powered by Gemini Vision                   |

## Operations Screens (20)

| Screen                    | Description                                                            |
|---------------------------|------------------------------------------------------------------------|
| OperationsDashboardScreen | Alert hub: low stock, expiring docs, vehicle service, filters by group |
| VehiclesScreen            | Vehicle management, mileage, service history, insurance expiry         |
| PantryScreen              | Inventory with expiry dates, minQty alerts, auto-restock to shopping   |
| MealPlanningScreen        | Weekly meal planner, generate shopping list from meals                 |
| DocumentsScreen           | Secure family document vault with category and expiry tracking         |
| EmergencyScreen           | Emergency contacts, procedures, first-aid references                   |
| EmergencyModeScreen       | Full-screen SOS broadcast to all family members                        |
| RewardsScreen             | Rewards catalog — kids redeem points                                   |
| AutomationScreen          | Household automation rules: 'when X, do Y'                             |
| MarketplaceScreen         | Post chores for kids to claim for bonus allowance                      |
| TimeEconomyScreen         | Time-blocking, productivity, daily schedule tracking                   |
| SmartHomeScreen           | Smart device dashboard, scenes, routines                               |
| ShoppingListScreen        | Collaborative shopping list with pantry-based suggestions              |
| RecipesScreen             | Family recipe collection with ingredient scaling                       |
| TravelPlanningScreen      | Trip planner with Plaid travel charge auto-import                      |
| PetTrackerScreen          | Pet profiles, vet visits, Plaid vet charge detection                   |
| HomeMaintenanceScreen     | Seasonal suggestions, recurring task auto-spawn                        |
| ChildcareManagerScreen    | Caregiver profiles, hours, auto-pay calculation, Plaid detection       |
| HomeInventoryScreen       | Home contents list, Plaid retail purchase suggestions                  |
| CarpoolManagerScreen      | Carpool routes with family member participant picker                   |

## AI Assistant Screens (6)

| Screen                      | Description                                                                 |
|-----------------------------|-----------------------------------------------------------------------------|
| AIAssistantScreen           | General AI chat with full family context, dynamic chips, AI action pipeline |
| AIMemoryScreen              | Memory insights, auto-milestone detection from goals and birthdays          |
| ParentingCoachScreen        | Expert parenting chat with child ages, screen time, memories injected       |
| NegotiatorScreen            | Conflict mediator with dynamic Chief of Staff briefing from real store data |
| DigitalTwinScreen           | 5 live family health scores computed from real store data                   |
| FamilySafetyAssistantScreen | Safety AI: child location, screen time, SOS, guardian context               |

## Health & Settings (6)

| Screen                  | Description                             |
|-------------------------|-----------------------------------------|
| HealthHubScreen         | Health overview for all family members  |
| MedicationManagerScreen | Medication schedules and dose reminders |
| SleepTrackerScreen      | Sleep duration and quality tracking     |
| WorkoutTrackerScreen    | Exercise logs and fitness goals         |

| Screen            | Description                                           |
|-------------------|-------------------------------------------------------|
| SettingsScreen    | App settings, profile, theme, language, notifications |
| HelpSupportScreen | Help documentation and support contact                |



## 4. State Management — 43 Zustand Stores

Every store is typed, persisted to MMKV, and follows the same pattern: state fields + action functions, no async thunks or middleware.

| Store                   | Key State                                                 | Key Actions                                             |
|-------------------------|-----------------------------------------------------------|---------------------------------------------------------|
| useAuthStore            | isAuthenticated, user, accessToken                        | login, logout, refreshToken                             |
| useAppStore             | isOnboarded, theme, locale                                | setOnboarded, setTheme                                  |
| useFamilyStore          | family, members, tasks, activeMemberId                    | addMember, updateMember, switchMember, addTask          |
| useFinanceStore         | bills, budgets, accounts, transactions, goals             | addBill, addTransaction, updateBudget, addGoal          |
| useGuardianStore        | devices, geofences, screenTimeRules, sosAlerts, approvals | addDevice, addGeofence, resolveSOSAlert, approveRequest |
| useOperationsStore      | vehicles, documents, pantryItems, tasks                   | addVehicle, addDocument, updatePantryItem               |
| useShoppingStore        | lists, items                                              | addItem, markPurchased, clearPurchased                  |
| useMemoryStore          | memories, tags                                            | addMemory, updateMemory, deleteMemory                   |
| useDebtStore            | debts, payoffPlans                                        | addDebt, recordPayment, syncFromPlaid                   |
| useNotificationsStore   | notifications, unreadCount                                | addNotification, markRead, clearAll                     |
| useAIStore              | conversations, chatHistory                                | sendMessage, clearHistory                               |
| useHomeMaintenanceStore | tasks, history                                            | addTask, completeTask (auto-spawns next recurring)      |
| usePetStore             | pets, vetVisits, events                                   | addPet, logVetVisit, addEvent                           |
| useTravelStore          | trips, itineraries                                        | addTrip, addItineraryItem, addExpense                   |
| useHomeInventoryStore   | items, categories                                         | addItem, updateItem, deleteItem                         |
| useCarpoolStore         | routes, participants                                      | addRoute, updateParticipants                            |
| useChildcareStore       | providers, bookings                                       | addProvider, addBooking                                 |
| useMoodStore            | logs, trends                                              | logMood, getMoodTrend                                   |
| useHabitsStore          | habits, completions, streaks                              | addHabit, recordCompletion, getStreak                   |
| useAllowanceStore       | allowances, payments                                      | setAllowance, recordPayment                             |
| useChoreStore           | chores, assignments, rotation                             | addChore, completeChore, rotateChores                   |
| useWealthStore          | assets, netWorth                                          | addAsset, updateNetWorth                                |
| useInsuranceStore       | policies                                                  | addPolicy, updatePolicy                                 |
| useUtilityStore         | bills, readings                                           | addBill, addReading                                     |
| useRecipesStore         | recipes, favorites                                        | addRecipe, toggleFavorite                               |
| useFamilyGoalsStore     | goals, milestones                                         | addGoal, updateProgress                                 |
| useJournalStore         | entries                                                   | addEntry, editEntry, deleteEntry                        |
| useTimelineStore        | events, milestones                                        | addEvent, addMilestone                                  |
| useBirthdayStore        | birthdays                                                 | addBirthday, getUpcoming                                |
| useGiftStore            | gifts, occasions                                          | addGift, markPurchased                                  |
| usePollsStore           | polls, votes                                              | createPoll, castVote                                    |
| useBucketListStore      | items                                                     | addItem, markComplete                                   |

| Store              | Key State                                                                     | Key Actions                        |
|--------------------|-------------------------------------------------------------------------------|------------------------------------|
| useAutomationStore | rules, conflicts                                                              | addRule, toggleRule                |
| useReputationStore | scores, history                                                               | awardPoints, deductPoints          |
| + 9 more           | Health, Homework, School, Legacy, Activities, Board, Meetings, Requests, Time | Standard add/update/delete actions |

## 5. Backend API Reference (57 Endpoints)

### Authentication

| Method | Endpoint       | Description                                                |
|--------|----------------|------------------------------------------------------------|
| POST   | /auth/register | Create user + family + first member in one transaction     |
| POST   | /auth/login    | Authenticate; returns access_token + refresh_token (JWT)   |
| POST   | /auth/refresh  | Exchange refresh token for new access + refresh token pair |

### Family

| Method           | Endpoint                      | Description                           |
|------------------|-------------------------------|---------------------------------------|
| GET              | /family/:id/members           | List all members                      |
| POST             | /family/:id/members           | Add a member                          |
| GET/PATCH/DELETE | /family/:id/members/:memberId | Get / update / remove a member        |
| GET/POST         | /sync/family                  | Fetch or upsert complete family state |
| POST             | /sync/queue                   | Persist a SyncAction event            |

### Guardian & Parental Controls

| Method                | Endpoint                               | Description                                          |
|-----------------------|----------------------------------------|------------------------------------------------------|
| POST                  | /guardian/devices/pair                 | Pair device with 6-char code                         |
| PATCH                 | /guardian/devices/:id/status           | Update GPS, battery, mode from child app             |
| POST                  | /guardian/devices/:id/command          | Send command (lock/unlock/bedtime/school/web_filter) |
| GET                   | /guardian/devices/:id/commands/pending | Child app polls for commands                         |
| PATCH                 | /guardian/commands/:id/executed        | Mark command executed                                |
| GET/POST/PATCH/DELETE | /guardian/geofences[:id]               | CRUD geofence zones                                  |
| GET/POST/PATCH/DELETE | /guardian/screen-time[:id]             | CRUD screen time rules                               |
| POST                  | /guardian/app-usage                    | Batch upsert app usage from child                    |
| POST/GET              | /guardian/sos[:id/resolve]             | Create SOS alert / resolve                           |
| GET/POST/PATCH        | /guardian/approvals[:id/respond]       | Approval request workflow                            |
| POST                  | /guardian/devices/:id/geofence-event   | Geofence entry/exit event + FCM push                 |
| GET/POST              | /guardian/devices/:id/messages         | Parent-child messages                                |

### Plaid Banking

| Method | Endpoint                 | Description                                          |
|--------|--------------------------|------------------------------------------------------|
| POST   | /plaid/create-link-token | Create Plaid Link token for mobile                   |
| POST   | /plaid/exchange-token    | Exchange public token; trigger initial sync          |
| GET    | /plaid/transactions      | Paginated transactions with search + category filter |
| GET    | /plaid/accounts          | Connected accounts with live balances                |

| Method | Endpoint                      | Description                                |
|--------|-------------------------------|--------------------------------------------|
| POST   | /plaid/sync                   | Pull fresh data for all connected accounts |
| DELETE | /plaid/items/:id              | Disconnect bank account                    |
| GET    | /plaid/spending/by-category   | Category totals for a given month          |
| GET    | /plaid/spending/monthly       | 6-month spending trend data                |
| GET    | /plaid/detected-subscriptions | AI-detect recurring subscriptions          |

## Plaid Autofill (9 endpoints — zero manual data entry)

| Endpoint                            | Populates                                        |
|-------------------------------------|--------------------------------------------------|
| /plaid/autofill/bills               | Bills screen — recurring monthly charges         |
| /plaid/autofill/budget-suggestions  | Budget screen — 3-month category averages        |
| /plaid/autofill/utilities           | Utility Tracker — detected utility payments      |
| /plaid/autofill/insurance           | Insurance Manager — detected policy payments     |
| /plaid/autofill/investment-accounts | Wealth Builder — investment accounts             |
| /plaid/autofill/travel-charges      | Travel Planner — flights, hotels, transport      |
| /plaid/autofill/retail-purchases    | Home Inventory — large retail purchases (>\$100) |
| /plaid/autofill/pet-charges         | Pet Tracker — vet and pet store charges          |
| /plaid/autofill/caregiver-payments  | Childcare Manager — Venmo/Zelle to caregivers    |

## AI Endpoints

| Endpoint            | AI Persona        | Context Sent                                                   |
|---------------------|-------------------|----------------------------------------------------------------|
| /ai/chat            | Family Assistant  | Members, bills, pantry, vehicles, tasks, SOS + ACTION pipeline |
| /ai/parenting-coach | Parenting Coach   | Child names + ages + screen time limits + recent memories      |
| /ai/negotiator      | Conflict Mediator | Family context + selected scenario scripts                     |
| /ai/digital-twin    | Family Analyst    | Finance scores, task rate, vehicle/pantry data                 |
| /ai/memory-insights | Memory Analyst    | Up to 20 family memories                                       |
| /ai/guardian-chat   | Safety Assistant  | Devices, SOS count, approvals, screen time rules               |
| /receipts/scan      | Vision (Gemini)   | Receipt image — extracts merchant, amount, items, category     |

## 6. Database Models

| Model                 | Key Fields                                                                                                               | Notes                                                    |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| User                  | id, email, passwordHash, familyId, role (ADMIN/MEMBER/VIEWER)                                                            | Auth identity; separate from FamilyMember profile        |
| Family                | id, name, timezone, currency                                                                                             | Top-level tenant; all data scoped here                   |
| FamilyMember          | id, familyId, name, role, email, status, points, level, isAdmin, avatarColor, linkedUserId, isPinProtected               | Profile entity; linkedUserId connects to User            |
| SyncEvent             | id, familyId, entity, action, payload (JSON), processedAt                                                                | Flexible event log; stores all app state changes as JSON |
| ChildDevice           | id, familyId, memberId, deviceName, platform, status, batteryLevel, latitude, longitude, pairingCode, isPaired, fcmToken | One row per paired child device                          |
| GeofenceZone          | id, familyId, name, latitude, longitude, radius, action, icon, color, isActive                                           | Zones for entry/exit alerts                              |
| ScreenTimeRule        | id, familyId, memberId, dailyLimitMinutes, blockedApps[], scheduledDowntime (JSON), isActive                             | Per-child screen time policy                             |
| SOSAlert              | id, familyId, memberId, deviceId, latitude, longitude, address, isResolved                                               | Emergency alerts with GPS                                |
| ParentApprovalRequest | id, familyId, memberId, type, title, description, status (pending/approved/denied)                                       | Child-to-parent approval workflow                        |
| GuardianCommand       | id, familyId, deviceId, type, payload (JSON), sentAt, executedAt, status                                                 | Command queue for child devices                          |
| AppUsageEntry         | id, deviceId, familyId, appName, packageName, usageMinutes, date                                                         | Daily app usage from child device                        |
| PlaidItem             | id, familyId, itemId, accessToken, institutionId, cursor, lastSyncedAt                                                   | One row per connected bank                               |

## 7. Integrations & Native Modules

| Integration               | Purpose                                                            | Security Notes                                                       |
|---------------------------|--------------------------------------------------------------------|----------------------------------------------------------------------|
| Plaid v42                 | Bank connection, transactions, balances, autofill                  | Access tokens stored server-side only; never in mobile               |
| Firebase FCM (server)     | Push parental commands, geofence alerts, SOS to devices            | Firebase Admin private key: server env var only, never in app bundle |
| Google Vertex AI          | 6 AI chat personas + Gemini Vision receipt scanning                | API key: server-side only                                            |
| expo-location             | Background GPS in mobile app; foreground + background in child app | Requires always-on permission on child device                        |
| expo-camera               | Receipt photo capture                                              | Camera permission, image processed server-side                       |
| expo-notifications        | Receive + display FCM push notifications                           | FCM token registered on login                                        |
| expo-auth-session         | Google OAuth sign-in                                               | PKCE flow, no client secret in app                                   |
| expo-apple-authentication | Apple Sign-In on iOS                                               | Following Apple's guidelines for first-party auth                    |
| expo-secure-store         | Encrypted JWT + biometric credential storage                       | AES encryption, hardware-backed on supported devices                 |
| expo-local-authentication | Face ID / Touch ID biometric sign-in                               | Credentials verified locally; no biometric data sent                 |
| react-native-mmkv         | Fast encrypted Zustand store persistence                           | MMKV encrypted with device key                                       |
| Android VPN (child)       | DNS-level web content filtering                                    | VPN runs locally on device; no traffic leaves to external VPN server |

## 8. Family Guardian Child App

A separate lightweight Android React Native app installed on children's devices. Always-on background services with minimal UI.

### Screens & Services

| Component             | Type             | Description                                                                                      |
|-----------------------|------------------|--------------------------------------------------------------------------------------------------|
| PairingScreen         | Screen           | Enter 6-digit code to link with parent family (first launch only)                                |
| HomeScreen            | Screen           | Child's view: status, chores, Today's app usage (top 5), SOS button, messages                    |
| MessagesScreen        | Screen           | Parent-child chat — polls 15s, child blue-right, parent gray-left                                |
| RestrictedScreen      | Screen           | Full-screen lock for lock/bedtime/school mode                                                    |
| LocationService       | Background       | GPS every 50m/1min; reverse geocode; geofence zone check on every update                         |
| CommandPoller         | Background       | Polls pending commands; handles lock/unlock/bedtime/school/web_filter/geofence_update/usage_sync |
| ScreenTimeService     | Background       | Reads Android UsageStatsManager; formats 'Xh Ym'; syncs to server every 2 heartbeats             |
| WebFilterService      | Background       | Manages VPN service: start/stop, update domain blocklists, handle permission request             |
| GuardianVpnService.kt | Native (Android) | Full VPN: TUN interface, DNS intercept on port 53, block by domain/keyword, NXDOMAIN response    |

| Component          | Type          | Description                                                                                  |
|--------------------|---------------|----------------------------------------------------------------------------------------------|
| WebFilterModule.kt | Native bridge | RN bridge: startWebFilter, stopWebFilter, isRunning, setBlockedDomains, requestVpnPermission |
| AppUsageModule.kt  | Native bridge | RN bridge: hasPermission, requestPermission, getUsageStats(hours), getCurrentApp             |

## 9. AI Features

### Context Builder (useFamilyContextString hook)

Injected into every AI request. Built fresh on each message from live store state:

| Context Field                           | Source                                          |
|-----------------------------------------|-------------------------------------------------|
| Family name + member names, roles, ages | useFamilyStore (ages computed from dateOfBirth) |
| Bills due in 14 days + overdue bills    | useFinanceStore.bills                           |
| Low pantry items (qty <= minQty)        | useOperationsStore.pantryItems                  |
| Vehicles with service due in 30 days    | useOperationsStore.vehicles                     |
| Overdue tasks                           | useFamilyStore.tasks                            |
| Unresolved SOS + pending approvals      | useGuardianStore                                |

### AI Action Pipeline

The backend appends instructions to Gemini's system prompt to emit structured ACTION blocks. When detected in the response, the mobile app strips the ACTION from display text and shows a confirmation banner:

```
ACTION:{"type":"add_task","payload":{"title":"Call doctor","dueDate":"2026-07-01","priority":"medium"}}
```

| Action Type | Store Dispatched                        |
|-------------|-----------------------------------------|
| add_task    | useFamilyStore.addTask(payload)         |
| add_bill    | useFinanceStore.addBill(payload)        |
| add_expense | useFinanceStore.addTransaction(payload) |
| add_memory  | useMemoryStore.addMemory(payload)       |

### Digital Twin — Live Family Health Scores

| Dimension        | Formula                                                                                             |
|------------------|-----------------------------------------------------------------------------------------------------|
| Financial Health | Base 50 + has positive balances (+20) + income > expenses (+30) — overdue bills (-10 each, max -30) |
| Productivity     | Task completion rate x 100 — overdue task penalties (-5 each, max -25)                              |
| Safety           | Base 70 — unresolved SOS (-20 each, max -40) — pending approvals (-5 each, max -20)                 |
| Home             | Base 80 — overdue vehicle service (-15 each) — low pantry items (-3 each, max -20)                  |
| Wellness         | Base 60 + screen time rules per child (+10 each), capped at 100                                     |
| Overall          | Average of all 5 dimensions, rounded to nearest integer                                             |



## 10. Key Features Summary

| #  | Feature                | Description                                                                                    |
|----|------------------------|------------------------------------------------------------------------------------------------|
| 1  | Multi-role family      | 5 roles (parent/guardian/child/grandparent/caregiver) with screen-level permission enforcement |
| 2  | Plaid banking          | Connect accounts; auto-sync transactions; detect bills, subs, utilities automatically          |
| 3  | AI action pipeline     | Conversational actions: 'add Netflix to bills' creates ACTION; store updated on confirm        |
| 4  | Receipt scanning       | Photo receipt; Gemini Vision extracts merchant/amount/items/tax; auto-dispatched               |
| 5  | Smart autofill         | 9 Plaid autofill endpoints — eliminate manual entry across finance and operations              |
| 6  | Parental controls      | Location, geofencing, screen time, app blocking, lock/unlock via FCM commands                  |
| 7  | VPN web filtering      | Android VPN intercepts DNS; blocks adult/violence/gambling — no root required                  |
| 8  | Real-time geofencing   | Named zones; entry/exit triggers FCM push: 'Aiden entered School Zone'                         |
| 9  | SOS alerts             | Child triggers SOS; GPS captured; FCM to all parents; resolved via Guardian dashboard          |
| 10 | Parent-child chat      | Persistent messaging via SyncEvents; polls 15s in child app                                    |
| 11 | Debt payoff planner    | Snowball/Avalanche with Plaid balance sync and month-by-month projection                       |
| 12 | Chore + allowance      | Auto-rotating chores, completion tracking, allowance linked to points                          |
| 13 | Gamification           | Points, levels, reputation, achievements, kids mode with large-icon UI                         |
| 14 | Spending insights      | Category breakdowns + 6-month trends from Plaid transaction data                               |
| 15 | Meal planning + pantry | Weekly planner, expiry/minQty, auto-add to shopping at qty=0                                   |
| 16 | Home maintenance       | Seasonal suggestions, recurring task auto-spawn on completion                                  |
| 17 | Travel planning        | Trip planner with Plaid travel charge auto-import                                              |
| 18 | Digital twin scores    | 5 live dimensions computed from real store data — no hardcoded values                          |
| 19 | Legacy vault           | Grandparent-accessible family history, milestone auto-creation from finance goals              |
| 20 | Biometric sign-in      | Face ID/Touch ID shortcut using stored credentials in SecureStore                              |

## 11. Folder Structure

### family-command-center-mobile/

```
src/
  config/          API URL, feature flags
  navigation/      AppNavigator, TabNavigator, 6 stack navigators, CustomTabBar
  screens/         100 screens in 9 folders
    auth/          3 screens (SignIn, SignUp, ForgotPassword)
    onboarding/    8 screens (Splash through GoalsSetup)
    dashboard/     5 screens
    family/        31 screens + guardian/ (7 screens)
    finance/       14 screens
    operations/    20 screens
    health/        4 screens
    ai/            6 screens
    settings/      2 screens
  services/        plaidService, autoFillService, debtService, ...
  store/           43 Zustand stores
  theme/           colors, spacing, ThemeContext
  types/           index.ts – all TypeScript interfaces
  utils/           buildFamilyContext, formatters
  i18n/           Translation files (en, fr, es, ...)
  android/         Native Android project
  ios/             Native iOS project
  docs/           This PDF
```

### family-command-center-api/

```
src/
  routes/          auth.ts, family.ts, guardian.ts, plaid.ts,
                  plaid-autofill.ts, debts.ts, receipts.ts, ai.ts, admin.ts
  middleware/      auth.ts (JWT verify + role check)
  utils/           firebase.ts, gemini.ts, prisma.ts
  prisma/
    schema.prisma  12 database models
```

### family-guardian-child/

```
src/
  screens/         HomeScreen, MessagesScreen, PairingScreen, RestrictedScreen
  services/        LocationService, CommandPoller, ScreenTimeService, WebFilterService
  api.ts           API calls with auth token
  storage.ts       SecureStore helpers
  android/.../child/
    AppUsageModule.kt, ChildPackage.kt, MainApplication.kt
  android/.../child/vpn/
    GuardianVpnService.kt, WebFilterModule.kt
```

## Summary

**Family Command Center** is a production-ready, full-stack family management platform covering every aspect of modern family life:

- **100 screens** across 9 functional areas
- **43 Zustand stores** with MMKV persistence — zero data loss on app restart
- **57 API endpoints** across auth, family, guardian, Plaid, AI, and admin
- **12 PostgreSQL database models** via Prisma ORM
- **6 AI personas** powered by Google Gemini with live family context injection
- **9 Plaid autofill endpoints** to eliminate manual finance data entry
- **Full parental controls**: GPS, geofencing, VPN filtering, screen time, SOS alerts
- **Parent-child messaging**, approval workflows, and remote device lock/unlock
- **Gamification**: points, levels, reputation, achievements for all family members
- **Role-based access**: every screen enforces permissions for parent/child/grandparent/caregiver